

Large-Scale Data Collection in Fieldwork Contexts: A Case Study of Clause Embedding in Zulu

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Abstract Some research questions require large-scale acceptability judgment datasets to answer them. Constructing such datasets can be challenging when the researcher has limited familiarity with the language. We investigate how to optimally design acceptability judgment collection protocols in such contexts. Taking clause embedding in Zulu as a case study, we collect acceptability judgments using protocols that manipulate a combination of setting, modality, prior exposure, and question format and measure how much data collected using these different protocols converges in the acceptability it implies. We find that unguided surveys can recover the same acceptabilities as elicitation, but need roughly three times as much data—up to about ten times at the highest precision—to match it. We attribute this cost to protocol-induced variability: elicitation, and group elicitation in particular, buys the most acceptability per judgment.

Keywords: acceptability judgments; field work; complementation; elicitation; data collection methodology; Zulu

1 Introduction

Linguistic acceptability judgments are typically collected using two main methods: elicitation and unguided survey tasks (Schütze & Sprouse 2014; Bard et al. 1996). One crucial difference between them lies in the degree of control they afford the researcher. On the one hand, elicitation allows a researcher to tailor judgment collection based both on the judgments themselves as well as auxiliary metalinguistic judgments and discussion. This interaction allows potential sources of ambiguity or misunderstanding to be identified and resolved as they arise; but it has the downside that it is hard to scale in cases where such scale is useful—e.g. when attempting to draw generalizations about broad swaths of a language’s lexicon.

On the other hand, unguided surveys support scalability (White & Rawlins 2016; 2020), but they do not provide the same level of control that elicitation does because participants generally complete tasks independently, without the possibility of clarification or adjustment during data collection. As a result, unguided surveys leave greater room for confounding factors, including misinterpretation of stimuli, task effects, and variability

in how participants approach the judgment task (Wasow & Arnold 2005; Gibson & Fedorenko 2010; 2013; Sprouse 2011; Sprouse et al. 2013; Sprouse & Almeida 2013).

The relative suitability of these two methods is closely tied to the amount of prior knowledge the researchers have about the language under investigation. For well-studied languages, especially those spoken by the researchers, many potential confounds can be anticipated and controlled for in advance, during stimulus construction. In such cases, the lack of real-time control in unguided surveys is less problematic.

In contrast, lesser-studied languages typically require substantially greater researcher involvement during data collection; and the degree to which researcher involvement is required depends on the researcher’s familiarity with the language and its properties. That is, there is an inverse relationship between language familiarity and the degree of experimental control required for reliable data collection (see Linzen & Oseki 2018).

language familiarity ↓ ⇒ required experimental control ↑

This inverse relationship has important consequences for research on lesser-studied languages. Because familiarity is limited, elicitation remains the primary and most reliable method of data collection. Unguided surveys, while occasionally used, pose substantial challenges in such contexts due to the increased risk of confounded judgments and the reduced ability to diagnose them. This creates an asymmetry with well-studied languages, for which both elicitation and unguided surveys can be used successfully.

This situation leads to an important methodological challenge: when a research question simply cannot be answered without large amounts of data—e.g. questions that pertain to the distributional or inferential behavior of large numbers of open class items (White & Rawlins 2016; 2020)—how should that data be collected?

To address this question, we assess the conditions under which judgments gathered using elicitation and judgments gathered using unguided surveys agree most. As a case study, we focus our investigation on clause embedding in Zulu, for reasons discussed in Section 2.

To conduct this investigation, we collected acceptability judgments for a shared set of Zulu sentences using a range of data-collection protocols. The protocols varied along three dimensions. First, they differed in modality, namely elicitation versus online surveys. The elicitation protocols further varied by setting, with participants either meeting one-on-one with the researcher or participating in group sessions. The online survey protocols, in turn, varied according to whether participants had previously taken part in an elicitation study. In each protocol, we systematically varied the way in which participants were asked to provide a judgment, varying both the nature of the response scale participants provided judgments on and whether or not participants receive an English translation against which to judge the Zulu sentence. We discuss these manipulations and their rationale in Section 3.

Our broader aim in pursuing this investigation is to provide concrete recommendations for how to design one’s approach to acceptability judgment collection in a context where scale is required (because the question of interest demands it) but language familiarity is low. We first establish, using only model-free measures of inter-annotator agreement, how much judgment variability the different protocols produce on the same set of expressions (Section 4). To quantify how much of that variability survives into our inferences about acceptability, we then develop a statistical model that synthesizes the data col-

lected under all protocols into (i) a single acceptability value; and (ii) a mapping, for each protocol, from that value to the response scale that protocol uses (Section 5)—yielding a proxy for the ‘ground truth’ acceptability of each sentence against which the acceptability inferred from any single protocol can be compared. Finally, we use this model to ask how little data suffices to recover our conclusions, by fitting it to subsets of the data and comparing with the full-data fit (Section 6).

Overall, we find that unguided surveys can be used to produce similarly precise estimates of acceptability but that they require at least three times as much data to produce estimates of similar precision. We attribute this finding to the substantially higher levels of variability we observe in judgments given in unguided surveys compared to elicitation. We furthermore observe that such variability reliably decreases in group elicitation sessions compared to one-on-one sessions. This pattern suggests that, while all protocols will produce the same average acceptability for a given set of expressions eventually (i.e. given enough data), elicitation in general (and group elicitation in particular) will produce precise acceptability estimates with substantially less data.

2 Topic of study

This study investigates clausal embedding in Zulu, focusing on the selection of clausal complements by matrix predicates. Complement selection shows both verb-specific behavior and clear cross-linguistic tendencies (Grimshaw 1979; 1990; 1994; Pesetsky 1982; Dowty 1991; Resnik 1996; Gleitman 1990; Fisher et al. 1991; Pinker 1994); see White (2015) for a review. Individual predicates differ in what kinds of clauses they allow, but these differences are not arbitrary. For example, attitude predicates typically select fully finite clauses, whereas implicative verbs (e.g. *begin*) tend to take smaller clausal complements, such as infinitives (Givón 1980; Karttunen 1971; Bolinger 1968; Ramchand & Svenonius 2014; Wurmbrand 2014; Wurmbrand & Lohninger 2023; Diessel & Tomasello 2001; Simons 2007; Hacquard & Wellwood 2012; Anand & Hacquard 2013; White et al. 2018; 2014; White & Rawlins 2018). At the same time, the shape of these generalizations depends on the grammatical resources of the language. If a language lacks a particular clause type, such as the subjunctive clause, it will use other available constructions (e.g. infinitives) in the semantic contexts where other languages would use subjunctives.

Most work on these issues has focused on Indo-European languages, which tend to have relatively small inventories of clause types, typically three or four. Zulu differs in this respect, with at least six distinct clausal constructions. This makes it possible to ask how the familiar patterns of complement selection are distributed across a richer set of syntactic options, and whether they extend in a straightforward way or reveal new distinctions.

The primary aim of this paper is methodological rather than theoretical: we focus on the collection and analysis of acceptability judgments, rather than on developing a detailed analysis of Zulu complementation. Nevertheless, understanding the scope and complexity of the empirical domain is essential for motivating the methodological choices discussed in the remainder of the paper. Accordingly, this section introduces the relevant constructions in Zulu through illustrative examples and highlights the parsing ambiguities that complicate the collection and interpretation of acceptability judgments.

2.1 Six types of embedded clauses in Zulu

We distinguish six types of embedded clauses in Zulu, listed in (1). Each of these clause types can occur as the complement of a matrix predicate, rather than (only) in adjunct contexts.

- (1) Syntactic clause types in Zulu
 - a. Indicative clauses
 - b. Subjunctive clauses
 - c. Small subjunctives
 - d. Infinitives
 - e. Gerunds
 - f. Participles

Most of these clause types have been recognized in the previous literature (Slattery 1981; Ziervogel et al. 1985; Van der Spuy 2001; Buell 2005; Zeller 2006; Halpert 2015; Carstens 2025), although we are not aware of any work that systematically compares all six. In this section, we introduce each clause type in turn, focusing on the properties that distinguish it from the others and that are relevant for the present study.

Indicative clauses are CPs introduced by the complementizer *ukuthi*. They exhibit the full range of tense contrasts found in matrix clauses and, like matrix clauses, allow preverbal agreeing subjects. Typical predicates that select indicative clauses include *cabanga* ‘think’ (2a) and *azi* ‘know’ (2b).

- (2) Indicative CP
 - a. Ngi-cabanga [ukuthi uZodwa u-zo-pheka].
 1SG.S-think COMP 1Zodwa 1.S-FUT-cook.IND
 ‘I think that Zodwa will cook.’
 - b. UZodwa u-y-azi [ukuthi u-win-ile].
 1Zodwa 1.S-DSJ-know COMP 1.S-win-PST.IND
 ‘Zodwa knows that she won.’

Subjunctive clauses resemble indicative clauses in that they are introduced by a complementizer and allow preverbal subjects. Unlike indicative clauses, however, they do not exhibit morphological tense marking. They also differ in their subject agreement paradigm, using a distinct allomorph for class 1 agreement (*a-* rather than *u-*), and in the form of the verbal final vowel, which is invariably *-e*.

- (3) Subjunctive CP
 - a. UZodwa u-kheth-e [ukuthi a-hamb-e endlini].
 1Zodwa 1.S-choose-PST COMP 1.S-go-SBJV house.LOC
 ‘Zodwa chose to go home.’
 - b. Ku-mele [ukuthi uZowda a-phek-e].
 15.S-must COMP 1Zodwa 1.S-cook-SBJV
 ‘Zodwa must cook.’

Several observations suggest that subjunctive clauses are structurally smaller than indicative clauses. First, they use the same negation marker as infinitives. It has been argued that in closely related Ndebele, this type of negation marking is a consequence of a reduced clause size (Pietraszko 2018a; 2021). Second, subjunctive clauses appear to be more permeable to raising: most hyperraising constructions in Zulu involve subjunctive rather than indicative complements (see, e.g., Zeller 2006; Halpert 2015). Finally, unlike indicative clauses, subjunctive clauses in Zulu have been reported to be transparent for NPI licensing (Carstens 2025). Similar asymmetries between indicative and subjunctive clauses have been observed in many other languages (see, among others, Hooper & Thompson 1973; Progovac 1993; Giannakidou & Quer 1997; Giannakidou 1998; Haegeman 2010; Wurmbrand & Lohninger 2023).

A lesser-known type of subjunctive complement is what we call the **Small Subjunctive**. This construction has previously been described for Zimbabwean Ndebele (Pietraszko 2017; 2018b; 2020), and our research indicates that it is also available in Zulu. We adopt the term *Small Subjunctive* (from Pietraszko 2017) because several diagnostics suggest that these complements are structurally smaller than subjunctive CPs. Unlike subjunctive CPs, they do not allow a complementizer (4a) or a preverbal subject (4b).

(4) Small Subjunctive

- a. UNandi u-ke [(**ukuthi*) a-phek-e].
 1Nandi 1.S-sometimes (**COMP*) 1.S-cook-SBJV
 ‘Nandi sometimes cooks.’
- b. **Ku-ke* [uNandi a-phek-e].
 15.S-sometimes 1Nandi 1.S-cook-SBJV
 ‘Nandi sometimes cooks.’

A further difference between subjunctive CPs and Small Subjunctives concerns tense morphology. While subjunctive CPs lack morphological tense marking, Small Subjunctives exhibit a two-way tense contrast: they appear in either a past or a non-past form, depending on the tense of the matrix predicate. This tense agreement is obligatory, as illustrated in (5).

- (5) a. Ng-a-buye [ng-a-pheka/**ngi-phek-e*].
 1SG.S-PST-later 1SG.S-PST-cook.SBJV/**1SG.S-cook-SBJV*
 ‘I cooked later.’
- b. Ngi-zo-buye [ngi-phek-e/**ng-a-pheka*].
 1SG.S-FUT-later 1SG.S-cook-SBJV/**1SG.S-PST-cook.SBJV*
 ‘I will cook later.’

Pietraszko (2017; 2018b) argues that this tense agreement is indicative of a reduced size of these embedded clauses. Subjunctive CPs are generally opaque to agreement, whereas the absence of the CP layer in Small Subjunctives—independently supported by the absence of a complementizer—allows tense agreement with the matrix clause.

The predicates that select Small Subjunctives typically express meanings encoded by adverbs in many other languages. Additional examples include *phinda* ‘again’, *ze* ‘finally’, *buye* ‘again/later’, *sheshe* ‘quickly’, and *jwayela* ‘usually’. Accordingly, these predicates have been referred to as *deficient verbs* or *adverbial auxiliaries* in previous literature (Slat-

tery 1981). Syntactically, however, they behave like raising verbs selecting reduced clausal complements, approximately of the size of VoiceP.

Infinitives in Zulu have the morphological form of verb nominalizations, a characteristic property of infinitives across Bantu languages (Schadeberg 2003). They consist of the verb stem preceded by the class 15 nominal prefix *ku-* and the augment vowel *u-* (6).

- (6) a. u-ku-dla
AU-15-food/eat
'food'; 'to eat'
- b. u-ku-pheka
AU-15-cook
'cooking; to cook'

The syntactic status of infinitives remains a matter of debate. While Halpert 2015 argues that they are externally DPs, other work analyzes them as TPs. This question is not directly relevant for the present discussion, beyond a similarity they share with gerunds, discussed below. Infinitives do not exhibit tense marking and, unlike CP complements, they do not license overt embedded subjects. Instead, clauses with infinitival complements are typically control constructions.¹ This illustrated by the verb *funa* 'want' in (7). When *funa* selects an infinitival complement, only the subject-control variant is possible, while an ECM structure with a distinct embedded subject is disallowed. The latter becomes possible only when the complement is a subjunctive CP (7c).

- (7) a. Ngi-funa uku-pheka.
1SG.S-want INF-cook
'I want to cook.'
- b. *Ngi-funa umama uku-pheka.
1SG.S-want 1mother INF-cook
'I want mother to cook.'
- c. Ngi-funa ukuthi umama a-phek-e.
1SG.S-want COMP 1mother 1.S-cook-SBJV
'I want mother to cook.'

Gerunds resemble infinitives in that they have the morphological form of class 15 nominalizations (8). Unlike infinitives, however, gerunds may have an overt subject expressed as a possessor (8b).

- (8) a. UZodwa w-a-lungiselela [uku-hamba ekuseni.]
1Zodwa 1.S-PST-arrange INF-go morning.LOC
'Zodwa arranged to leave in the morning.'
- b. UZodwa w-a-lungiselela [uku-hamba kwakhe.]
1Zodwa 1.S-PST-arrange GER-go 1.POSS
'Zodwa arranged his/her departure.'

¹ Halpert 2015 argues that infinitival complements in Zulu do not permit raising out of them, whereas Pietraszko 2019 shows that such raising is possible in Ndebele, despite infinitives in the two languages being otherwise ostensibly identical. We did not investigate this issue in this study, and our findings are compatible with either possibility.

Whether infinitives and gerunds constitute distinct clause types is an empirical question. One possibility is that they have different syntactic structures, with only gerunds licensing overt (possessive) subjects. Alternatively, they may instantiate the same underlying structure, with the terms *infinitive* and *gerund* simply describing whether a possessive subject is overtly realized. Halpert (2015; 2022) takes the latter view, assuming that infinitives in Zulu can have possessive subjects. Our findings, however, support the former view. Infinitives and gerunds exhibit different selectional distributions across predicates, suggesting they are in fact different clause types. For example, the predicate *jwayela* ‘usually’ combines with an infinitival complement but not with a gerund (9).

- (9) USipho u-jwayele [uku-pheka (*kwakhe).]
 1Zodwa 1.S-usually INF-cook (*1.POSS)
 ‘Sipho usually cooks.’

Figure 6 in Section 5 shows the distribution of infinitives and gerunds across the full set of predicates, revealing additional contrasts of this kind.

The final clause type is the **participle**. Morphologically, participles can be distinguished from the other clause types by their distinct class 1 subject agreement allomorph, namely *e-*. They are typically selected by perception verbs such as *bona* ‘see’ (10a) and *zwa* ‘hear’ (10b).

- (10) a. Ngi-bon-e uThabo e-hamba.
 1SG.S-see-PST 1Thabo 1.S-go.PTCP
 ‘I saw Thabo leave.’
 b. Ngi-zw-e uThabo e-hamba.
 1SG.S-hear-PST 1Thabo 1.S-go.PTCP
 ‘I heard Thabo leave.’

In constructing the experimental stimuli, we made extensive use of class 1 subjects because the distinct class 1 agreement morphology provides an unambiguous diagnostic for distinguishing indicative, subjunctive, and participial clauses. The relevant agreement allomorphs are summarized in (11).

	Clause type	Class 1 agreement prefix
(11)	Indicative	<i>u-</i>
	Subjunctive	<i>a-</i>
	Participle	<i>e-</i>

The six clause types discussed above define the empirical domain of the present study. A central challenge in investigating their distribution is that many of the relevant strings are structurally ambiguous. These ambiguities complicate the collection and interpretation of acceptability judgments and motivate several aspects of our experimental design. We discuss the main parsing confounds in the following subsection.

2.2 Parsing confounds

Many of the embedding constructions discussed above give rise to structural ambiguities, allowing the same surface string to receive more than one syntactic analysis. Such ambi-

guities pose a challenge for the collection of acceptability judgments because a sentence that is unacceptable under the intended parse may nevertheless receive a positive judgment under an unintended interpretation. The parsing confounds described below were encountered repeatedly during our fieldwork and informed the design of the elicitation and survey tasks described in Section 3. They may also be relevant for future research on complementation in Zulu and related languages.

CP parsed as an infinitive

The complementizer *ukuthi*, used in both indicative and subjunctive CPs, is historically a class-15 nominalization of the verb *thi* ‘say’. Because infinitives are likewise class 15 verbal nominalizations, the complementizer *ukuthi* is homophonous with the infinitival form of the verb *thi* ‘say’:

- (12) a. uku-thi
15/INF-say
‘to say’
b. ukuthi
COMP

Consequently, a string intended to contain an embedded CP may instead be parsed as an infinitival complement headed by the verb *thi* ‘say’. Consider the verb *zama* ‘try’, which selects infinitival complements (13a) but not indicative CPs (13b).

- (13) a. Ngi-zama uku-pheka.
1SG.S-try INF-cook
‘I am trying to cook.’
b. *Ngi-zama ukuthi ngi-ya-pheka.
1SG.S-try COMP 1SG.S-DSJ-cook.IND
‘I am trying to cook.’

The string in (13b), however, is acceptable under an alternative parse in which *ukuthi* is interpreted as the infinitival form of the verb *thi* ‘say’, which itself selects an indicative CP (14). This CP lacks an overt complementizer but complementizer drop is independently allowed in CPs selected by *thi*.

- (14) Ngi-zama uku-thi ngi-ya-pheka.
1SG.S-try INF-say 1SG.S-DSJ-cook.IND
‘I am trying to say that I am cooking.’

Throughout the remainder of the paper, we refer to the English translation corresponding to the intended parse as the *target translation*, and to translations corresponding to unintended parses as *non-target translations*. For example, the targeted translation of (13b) is “I tried to cook”, whereas the non-target translation is “I tried to say I am cooking”.

Gerund parsed as an infinitive

As discussed above, gerunds differ from infinitives in allowing an overt possessive subject. As illustrated in (9), the predicate *jwayela* ‘usually’ selects infinitival complements but

not gerunds. However, possessive DPs also admit a locative interpretation, as illustrated in (15).

- (15) Ngi-zo-pheka kwakhe.
 1SG.S-FUT-cook 1.POSS
 ‘I will cook at his/her place.’

This ambiguity gives rise to the following parsing confound. A string intended to contain a gerund may instead be parsed as an infinitive containing a locative phrase if the possessive DP is interpreted as a locative phrase rather than as the subject of the embedded clause. Consequently, a predicate that does not license gerund complements, such as *jwayela*, may nevertheless appear to do so, but the only acceptable parse is an infinitival one (16).

- (16) USipho u-jwayele [uku-pheka kwakhe.]
 1Sipho 1.S-usually INF-cook 1.POSS
 ‘Sipho usually cooks at his/her place.’
 Not: ‘Sipho usually cooks.’

Small Subjunctive parsed as a conjunct

A Small Subjunctive clause may be parsed not as the complement of the matrix predicate, but as the second conjunct in an event conjunction, where the two events are interpreted as occurring in temporal sequence (18). Since this is an ordinary coordination structure, it is, in principle, available with any predicate. Consequently, a predicate that does not select Small Subjunctive complements may nevertheless appear to do so under the coordination parse. This is illustrated below by the predicate *xolisa* ‘be sorry/apologize’, which selects infinitival complements (17) but not Small Subjunctives, as seen from the unavailability of the target translation in (18).

- (17) UZodwa w-a-xolisa uku-phazamisa.
 1Zodwa 1.S-PST-apologize INF-interrupt
 ‘Zodwa was sorry to interrupt.’
- (18) UZodwa w-a-xolisa w-a-phazamisa.
 1Zodwa 1.S-PST-apologize 1.S-PST-interrupt.SBJV
 ‘Zodwa apologized and (then) interrupted.’
 Not: Zodwa was sorry to interrupt.

Participle parsed as an adjunct

A participial clause intended as the complement of a matrix predicate may instead be parsed as a temporal adjunct, yielding a meaning roughly equivalent to an English *while*-clause. Since the latter interpretation involves adjunction, not complementation, a predicate that does not select participial complements may nonetheless be followed by a participle on the adjunct parse. This is illustrated again with the verb *xolisa* ‘apologize/be sorry’, which, as we saw in (17), selects infinitives. The embedding parse is unavailable when the this predicate is followed by a participle, and instead only the non-target parse of a temporal adjunct is available:

- (19) UZodwa w-a-xolisa e-phazamisa.
 1Zodwa 1.S-PST-apologize 1.S-interrupt.PTCP
 ‘Zodwa apologized while interrupting.’
 Not: Zodwa was sorry to interrupt.

The six clause types and the parsing confounds discussed above illustrate the empirical complexity of complement selection in Zulu. Together, they raise methodological questions about how to elicit judgments of predicate–frame acceptability—where we use *frame* to refer to an embedded clause type—while controlling for structural ambiguity. These questions extend beyond structural ambiguity, however, and include how to distinguish underlying selectional restrictions from variation in the naturalness or frequency of use of different predicate–frame combinations, as well as from preferences for one interpretation of an ambiguous sentence over another. The following section presents the data collection procedures we developed to address these questions.

3 Data collection

3.1 Experimental design

The data reported in this study were collected across four separate research studies conducted over a five-year period: (i) a pilot elicitation study, (ii) a large-scale elicitation study conducted during fieldwork in Durban, South Africa, and (iii–iv) two subsequent online survey studies.

These studies employed two distinct modalities of data collection: (i) guided elicitation, conducted through direct interaction between the researcher and participants, and (ii) unguided data collection, implemented via online questionnaires completed independently by participants.

In addition to the elicitation vs. online survey distinction, the studies also varied along a second design dimension, namely whether data were collected from participants individually or in group settings. The elicitation studies included both individual and group-based sessions (in the pilot study) or exclusively group-based sessions (in the Durban study), while the unguided online studies were completed individually by participants without interaction with other participants or the researcher.

Participant overlap varied across studies. The pilot and Durban studies involved entirely distinct participant pools. The first online study included some participants who had previously taken part in the Durban study. The second online study, in turn, drew heavily on participants from the first online study, resulting in substantial overlap between those two datasets. Table 1 summarizes the modality, settings and participants in the four studies.

This multi-study design allowed us to compare data obtained across different modalities (elicitation vs. online surveys) and response settings (individual vs. group). The partial overlap in participant pools across studies was needed to examine whether prior participation in guided elicitation influences participants’ responses in subsequent unguided tasks. In addition, the studies incorporated multiple question formats, all designed to probe acceptability but differing along three dimensions: i) the context in which the question was posed, ii) the instructions provided to participants prior to the answering the questions, and iii) the structure of the response scale. Despite these differences, all

Study	Modality	Setting	Participant n & overlap	# responses
Pilot	Elicitation	Individual & group	8 (all new)	2014
Durban	Elicitation	Group	84 (all new)	6707
Online 1	Online surveys	Individual	134 (44% overlap w Durban)	111791
Online 2	Online surveys	Individual	195 (56% overlap w Online 1)	221936

Table 1: Overview of studies: modality, response setting, and participants.

question formats were constructed to yield ordinal judgments, enabling systematic comparison across formats. This design therefore allows us to evaluate not only the effects of data collection modality and response setting, but also the extent to which different ways of formulating questions, and prior exposure to elicitation tasks, shape participants' responses. The following subsections describe each study in more detail.

3.2 Guided Elicitation Studies: Pilot and Durban Fieldwork

Two of the four studies – the pilot study and the Durban fieldwork study – employed guided elicitation. In both cases, data were collected through direct interaction between the researcher and participants, with the researcher presenting stimuli and guiding participants through a series of judgment tasks.

The elicitation studies were designed with the following goals in mind:

- (20) Goals (elicitation studies)
- Examining the effect of response setting (individual vs. group)**
We used both traditional one-on-one elicitation, as well as group sessions, allowing us to assess whether/how judgments differ depending on whether they are produced independently or in a group context that permits discussion.
 - Testing different question formats**
We piloted different ways of probing acceptability judgments prior to their use in online surveys.
 - Establishing a participant pool with elicitation experience**
We created a large group of participants who had prior experience providing acceptability judgments. This allowed us to later assess whether/how such experience affects participants' responses in unguided online surveys.
 - Expanding the data set for subsequent online studies**
These studies served to build an initial dataset of acceptability judgments that could be used to inform the design and interpretation of the subsequent online questionnaires.

3.2.1 Participants and Research Sites

Pilot Study

The pilot study was conducted at the University of Rochester. Participants consisted of 8 speakers of Zulu, including students at the University of Rochester and students at the University of KwaZulu-Natal (Durban, South Africa), who participated remotely. Among the participants, one was a bilingual speaker of Zulu and Siswati, and another was a

bilingual speaker of Zulu and Sesotho. One participant was from the Johannesburg area, while the remaining participants were from the Durban area. Participants ranged in age from 19 to 28.

Durban Fieldwork Study

The second elicitation study was conducted at the University of KwaZulu-Natal in Durban, South Africa. A total of 84 university students participated in the study. The participants had a mean age of 21.8 years (range: 17–27). Most (92.9%) reported growing up in KwaZulu-Natal, with the participants from Gauteng and Mpumalanga. The majority (71%) reported Zulu as their only childhood language, while others reported speaking Zulu alongside one or more other languages during childhood.

3.2.2 Pre-study Orientation

Prior to elicitation, participants were provided with an explanation of the nature of the judgments being requested. The purpose of this explanation was to minimize known sources of misunderstanding in acceptability judgment tasks.

Three key points were addressed. First, we illustrated the difference between descriptive and prescriptive correctness, and explicitly informed the participants that the study concerns how they themselves use the language, rather than whether a sentence conforms to prescriptive norms. Second, we drew participants' attention to dialectal and inter-speaker variation, emphasizing that differences in judgments are both normal and informative. This was intended to encourage participants in group settings to provide responses reflecting their own judgment rather than attempting to align with others. Third, participants were shown the distinction between a sentence that is descriptively well-formed and one that is merely interpretable. This clarification was important in avoiding a response strategy in which sentences are evaluated as acceptable simply because their intended meaning can be inferred. This additionally addressed another common pitfall in elicitation in which participants calibrate their judgments to what they expect the researcher, as a non-native speaker, to be able to produce.

3.2.3 Elicitation Procedures: Pilot

In the pilot elicitation study, 5 participants took part in individual sessions, and 3 participants took part in a single group session. Across both individual and group sessions, the elicitation procedure followed the same general format. Participants were presented with one Zulu sentence at a time and were asked to read it aloud. Immediately after reading the sentence, participants provided an acceptability judgment, and then an English translation of the Zulu sentence. This last step was needed to control for intended interpretation and potential confounds produced by lexical and structural ambiguity.

In group sessions, participants took turns reading, judging, and translating sentences. After one participant completed these steps, the remaining participants were invited to provide their own acceptability judgments and to indicate whether they agreed with the proposed translation. This format frequently led to discussion among participants, particularly in cases where judgments differed or where some participants identified interpretations that others had not initially recognized. In some instances, such discussion led participants to revise their original judgments after recognizing an interpretation that

had not been immediately apparent. In other cases, participants maintained their initial judgments, recognizing that others had judged the sentence differently.

We used two types of response scale in the pilot study: a binary scale and a 6-point numerical scale (0-5, with 5 being the highest acceptability). The two question formats are listed in (21). (The numbering of question formats reflects the history of the experimental design and should not be interpreted as meaningful).

(21) **Question Formats used in the pilot study**

Format 15

Is this a good isiZulu sentence?

[Zulu sentence]

Response options:

☐ Yes

☐ No

What does the sentence mean?

Format 16

Is this a good isiZulu sentence?

[Zulu sentence]

Response options:

☐ 5

∴ [intermediate points on a 6-point scale]

☐ 0

What does the sentence mean?

3.2.4 Elicitation Procedures: Durban Fieldwork

Data in the Durban study were collected through in-person elicitation conducted exclusively in group settings, with no individual sessions. 84 participants were divided into 8 groups, ranging in size from 7 to 12 participants.

As in the pilot, participants were presented with one Zulu sentence at a time and asked to respond to a question about each sentence. Participants were allowed to discuss their judgments with one another, to introduce contextual considerations, and to ask follow-up questions. After a brief discussion, participants provided their independent responses,

with the understanding that they need not align with others. Participants were not able to see the final responses submitted by the other members of the group.

The Durban study incorporated multiple question formats (QFs) to identify the most effective ways of eliciting acceptability judgments. We first classify the QFs according to whether the Zulu sentence was accompanied by an English translation. Consider first translation-free formats, listed in (22).

(22) **Translation-free Question Formats**

Format 1

Intro text: In this survey, you will be given isiZulu sentences. Please rate the acceptability of each sentence on a scale from 1 to 7, where 1 means that the sentence is completely broken, while 7 indicates a perfectly normal isiZulu sentence.

Is this a good isiZulu sentence?

[Zulu sentence]

Response options:

- ☐ 7 (Perfectly normal)
- ⋮ [intermediate points on a 7-point scale without a description]
- ☐ 1 (Completely broken)

Format 5

Intro text: In this survey, you will be given one isiZulu sentence at a time and will be asked to pick the most appropriate answer to the question “Is this something you might say?”

Is this something you might say?

[Zulu sentence]

Response options:

- ☐ Definitely yes
- ☐ Probably
- ☐ I’m not sure
- ☐ Probably not
- ☐ Definitely not

Format 6

Intro text: In this survey, you will be given one isiZulu sentence at a time and will be asked to pick the most appropriate answer to the question “Can you imagine yourself saying?”

Can you imagine yourself saying:

[Zulu sentence]

Response options:

- ☐ No problem
- ☐ Probably
- ☐ I’m not sure
- ☐ Probably not
- ☐ Definitely not

Format 7

Intro text: In this survey, you will be asked to pick the best statement describing an isiZulu sentence.

Which statement below best describes the sentence:

[Zulu sentence]

Response options:

- ☐ It’s correct and I can imagine saying it.
- ☐ It’s correct but I wouldn’t say it.
- ☐ I’m not sure if it’s correct or broken.
- ☐ It’s broken.

Format 8

Intro text: In this survey, you will see one isiZulu sentence at a time and will be asked to assess if it's something a native speaker of isiZulu would say. A native speaker of isiZulu is someone who learned the language in early childhood and used it as their main language at home. You are a native speaker of isiZulu.

Would a native speaker of isiZulu say:

[Zulu sentence]

Response options:

- ☐ No problem
- ☐ Probably
- ☐ I'm not sure
- ☐ Probably not
- ☐ Definitely not

The translation-free QFs varied along three main dimensions: (i) the wording of the prompt, (ii) the granularity of the response scale, and (iii) whether the response options were numerical or descriptive.

The wording of the prompt elicits acceptability judgments from slightly different perspectives. QF 1 adopts a sentence-centered perspective by asking whether the stimulus is a good isiZulu sentence. By contrast, QFs 5 and 6 adopt the speaker's perspective, asking participants to evaluate whether they themselves would be likely to produce the sentence (*Is this something you might say?*, *Can you imagine yourself saying this?*). Finally, QF 8 shifts the perspective from the individual speaker to the speech community (*Would a native speaker of isiZulu say...?*).

All QFs except QF 1 use a fully descriptive response scale. QF 1 differs in that only the endpoints of the scale are labeled ("Perfectly normal" and "Completely broken"), while the intermediate points are represented solely by numbers.

QF 7 was designed to test a more specific hypothesis, namely, that speakers may base acceptability judgments on use-related considerations, such as frequency of use or the likelihood of choosing one structure over another available alternative. To probe this possibility, the response scale included two options that both classified the sentence as acceptable, while explicitly distinguishing between acceptability and use.

In addition to the translation-free question formats, we included two QFs in which the Zulu sentence was accompanied by an English translation (listed in (23)).

(23) Translation-based Question Formats**Format 3**

Intro text: In this survey, you will be given isiZulu sentences with their English translation. You will be asked about both the acceptability of the isiZulu sentence and the correctness of the translation.

Which statement is true about the sentence:

[Zulu sentence]

[English translation]

Response options:

- ☐ The Zulu sentence is good and the translation is ok.
- ☐ The Zulu sentence is good but the translation is incorrect.
- ☐ The Zulu sentence sounds a bit odd but the translation is ok.
- ☐ The Zulu sentence sounds a bit odd and the translation is incorrect.
- ☐ The Zulu sentence is broken.

Format 11

Intro text: In this survey, you will be given isiZulu sentences with their English translations. You will be asked to assess their acceptability by choosing one of the provided options.

Is this a good isiZulu sentence?

[Zulu sentence]

[English translation]

Response options:

- ☐ It's perfectly normal.
- ☐ I'm not sure if I would say it, but it sounds correct.
- ☐ I'm not sure if it's correct or broken.
- ☐ I think it's broken.
- ☐ It's definitely broken.
- ☐ It's not broken but it means something different.

We introduced these translation-based formats because our pilot study revealed that both lexical and structural ambiguities are common and can confound conclusions about embedding patterns. By providing an English translation, we were able to elicit acceptability judgments under a specific intended interpretation, thereby controlling for these potential confounds.

3.3 Unguided online studies

The two online studies included a total of 222 participants with a mean age of 23.9 years (range: 17–60). Most participants (approximately 96%) reported growing up in KwaZulu-Natal, with the remainder primarily from Gauteng, Mpumalanga, and Limpopo. The majority (65.3%) reported Zulu as their only childhood language, while the rest reported speaking Zulu alongside one or more other languages during childhood.

The online studies were conducted using an online survey interface built with jsPsych (de Leeuw et al. 2023) and served using the JATOS platform (Lange et al. 2015). Both studies included question formats 1, 3, 5, 6, 7, 8, and 11 from the Durban study.

Surveys were composed of 45 sentences, including 5 calibration sentences and 40 target sentences. Each survey entirely consisted of a single question format and began with a page containing the corresponding instruction text (see (22)–(23)). In the first online study, participants were given 3 surveys per day, all from the same question format. 23 hours after completion of the 3 surveys, a participant was given access to 3 more surveys, from a different question format. In online study 2, we increased the surveys per day to 4 for each participant. This process repeated until each participant answered all 7 question formats tested, or until the end of the study period.

Since we had a large number of sentences, we found it important to stratify surveys in order to ensure an even distribution of good and bad expressions. For example, without stratification, a given random survey might accidentally contain 40 acceptable sentences, potentially causing participants to recalibrate their internal judgment scale.

To accomplish this, we used a preliminary version of the synthesizing model described in Section 5 fit to the elicitation data from the Pilot and Durban studies. For this predictive model, we included additional predictors of acceptability to achieve better generalization to the new expressions used in the online surveys as described in Appendix B. This model was used to perform posterior inference on all of the sentences, resulting in a predicted acceptability value for each sentence. We binned the sentences into 5 bins based on their inferred acceptabilities and sampled from the bins such that each set of 40 sentences contained an equal number from each.

For translation-based question formats (3 and 11), we additionally stratified the survey items based on whether they included a target or a non-target translation. Since we had significantly more target compared to non-target translations, we simply sampled at least one non-target translation item per survey.

4 Agreement across protocols

Before fitting any model to our data, we ask a more basic question: how far do the raw responses themselves agree, and how much does that agreement depend on the conditions under which we collect them? We use *protocol* throughout to mean a particular combination of a setting—individual or group—a modality—elicitation or online—a level of prior exposure, and a question format (QF), where a question format comprises both the prompt and the response scale together with its labels. Different protocols appear to pull the same speakers toward different responses, and we refer to the variability that this introduces as *protocol-induced variability*. Our aim in this section is to characterize protocol-induced variability using only model-free measures, so that the reader sees how much of it there is before meeting the model that we use to absorb it. We proceed in three steps: we first ask how far speakers agree and how that agreement varies with modality, setting, and prior exposure; we then ask whether two question formats measure the same thing for a given speaker, focusing on the reference translation; and we finally separate correctness from use, before placing every protocol on a common scale.

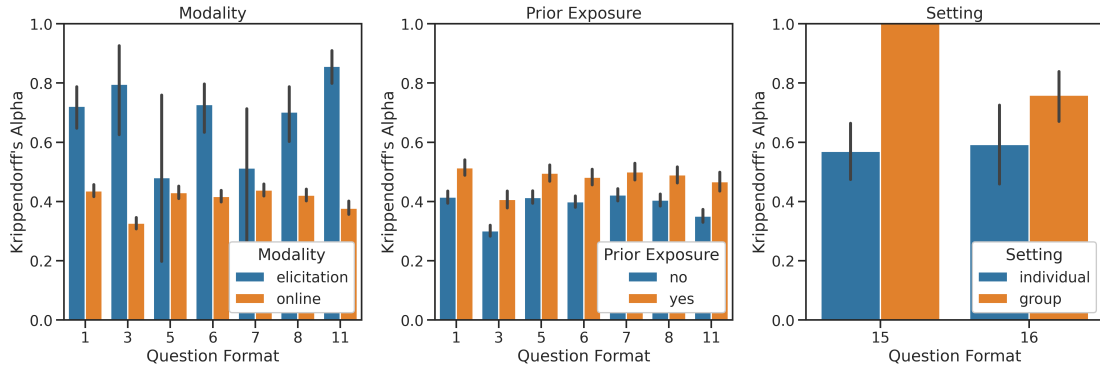


Figure 1: Inter-annotator agreement (Krippendorff's alpha, 95% percentile interval) between annotators, independently across question format, and additionally, independently across **Left:** modality (online vs. elicitation), **Center:** prior exposure, and **Right:** setting.

4.1 Agreement varies with modality, setting, and prior exposure

We begin with inter-annotator agreement (Artstein & Poesio 2008), which we quantify with Krippendorff's alpha (Castro 2017)—a measure of the extent to which annotators (i.e. elicitation or survey participants) converge in their ratings beyond what chance alone would produce. We compute alpha separately for each question format, on a speaker $\times$ sentence table of Zulu acceptability judgments, resolving the occasional duplicate cell by taking the median judgment so that each speaker-sentence pair contributes a single representative value. We then recompute this per-format agreement within each level of three factors: *modality* (elicitation vs. online), *setting* (group vs. individual), and *prior exposure* (whether an online participant had previously taken part in an elicitation session). The relevant studies and the question formats each one used are given in Table 2.

Study Name	Question Formats
Pilot	{15, 16}
Durban	{1, 3, 5, 6, 7, 8, 11}
Online1	{1, 3, 5, 6, 7, 8, 11}
Online2	{1, 3, 5, 6, 7, 8, 11}

Table 2: Study names and the question formats used in each.

Three patterns emerge, holding in the same direction across question formats (Figure 1): (i) agreement tends to be higher in elicitation than online surveys; (ii) higher in group sessions than in individual ones; and (iii) higher among participants with prior exposure to elicitation than among those without it. Each of these contrasts is visible across question formats, though each rests on a modest number of speakers, so we treat them as suggestive of how the protocol shapes convergence rather than as precise estimates of its size. This is the first sense in which the protocol matters: speakers agree with one another more when they are guided, when they judge together, and when they have taken part in elicitation before.

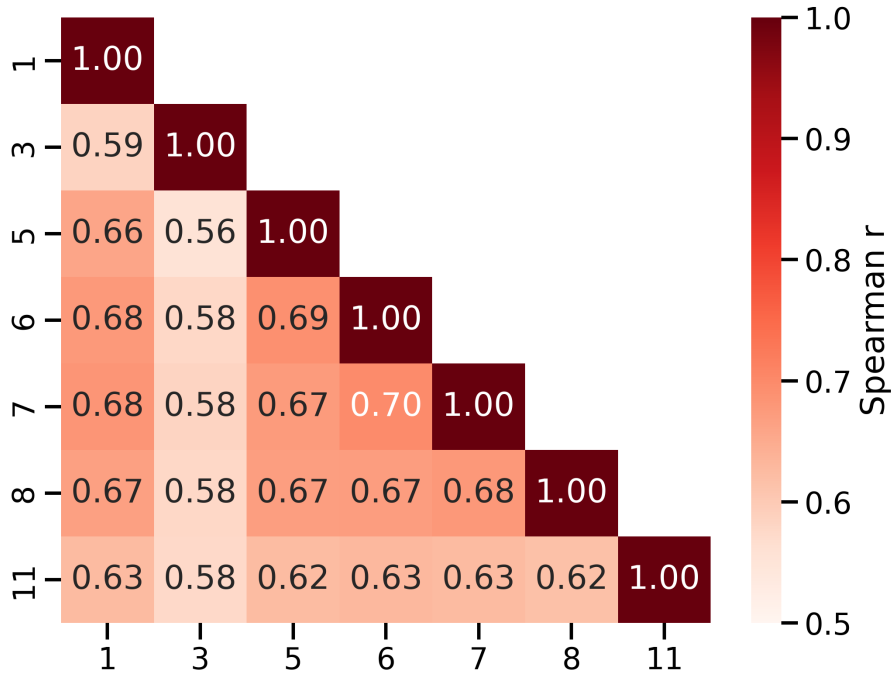


Figure 2: Spearman correlations between all pairs of question formats.

4.2 Including a reference translation shifts the Zulu-sentence judgment

Agreement tells us whether speakers converge; it does not tell us whether two question formats are measuring the same thing for a given speaker. To get at this, we turn to cross-format correlation. We compute Spearman’s rank correlation on the raw ordinal responses, and we compute it only between sentences that the same speaker judged under two different formats, so that each correlation reflects within-speaker, across-format consistency. For instance, the correlation of 0.68 between QF1 and QF7 reflects how consistently the same speakers ranked the same Zulu sentences under those two formats.

This lets us evaluate one design choice in particular: the reference translation. QFs 3 and 11 resemble the translation-free formats but pair each Zulu sentence with an English translation, with the intent of eliciting an independent judgment of the sentence and of the appropriateness of that translation; their ordinal coding is given in Appendix A. If a reference translation leaves the Zulu sentence judgment unaffected, QFs 3 and 11 should correlate with the translation-free formats about as well as those formats correlate with one another. They do not: across the board, both QF3 and QF11 correlate with the other formats less strongly than the translation-free formats correlate among themselves (Figure 2). Thus, a reference translation appears to affect the judgment a speaker gives of the Zulu sentence, rather than merely adding a second, separable judgment of the translation.

This pattern is somewhat surprising, since the entire point of including the reference translations is to guide participants toward the intended parse of the sentence in cases where, e.g., the Zulu sentence is acceptable on a non-target translation but not the target. So *a priori*, it might seem plausible that participants produce a judgment based on some aggregate over the possible parses—e.g. they might take a central tendency such as the median or they might respond based on the most acceptable parse (and thereby take the

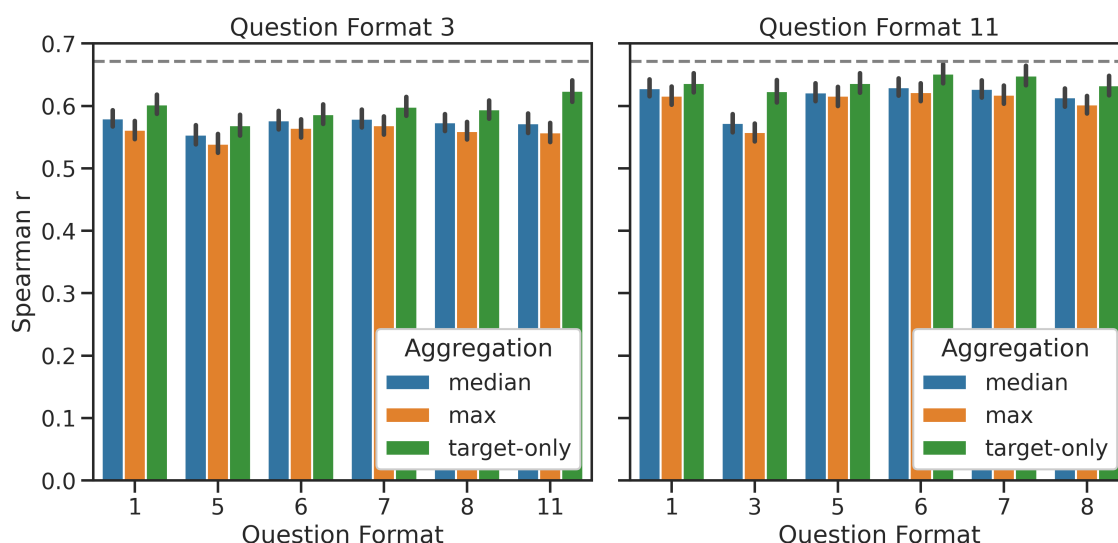


Figure 3: Left: Spearman correlation between QF3 and other QFs. Right: Same, for QF11. Dashed lines indicate the average correlation between QF1 and the other translation-free QFs. Errorbars indicate bootstrapped 95% percentile intervals.

max across translations). It turns out, however, that excluding the non-target translations raises the correlation between the translation-based formats and the translation-free ones (Figure 3). This pattern may suggest that participants in the translation-free protocols were actually judging acceptability relative to our target translations implicitly. We suspect that, insofar as they are doing this, it may be a product of them noticing that the expressions we tested tended to involve clause embedding and therefore, even if a particular sentence was compatible with an alternative parse, they should judge against the clause embedding parse.

4.3 Separating correctness from use

A second design choice targets a different worry: that a well-formed sentence may draw a low rating for reasons unrelated to well-formedness—register, frequency, or lexical choice, among others. QF7 addresses this by separating correctness from use, letting a speaker mark a sentence as correct while also indicating that they would not themselves say it. This gives us a way to check whether the low ratings a sentence receives elsewhere track its grammaticality or its usage.

We test this by comparing, for sentences judged by the same speakers in both formats, the QF1 ratings of sentences those speakers classified each way in QF7. Sentences a speaker rated “correct but wouldn’t say” in QF7 are rated significantly lower in QF1 than sentences the same speaker rated “correct and would say” (Figure 4, Mann-Whitney $N_1 = 1735, N_2 = 5239, U = 6.7 \times 10^7, p \approx 0$). This shows that use-related considerations are a significant component of a numerical response – speakers often judge a sentence as correct on a descriptive response scale, and give that sentence a low or middling numerical rating if the likelihood of use is low. The results therefore suggest that at least some of the spread in a numerical format such as QF1 reflects whether a speaker would use a sentence rather than whether they judge it as well-formed. This is a second way in which the choice of protocol feeds into the response.

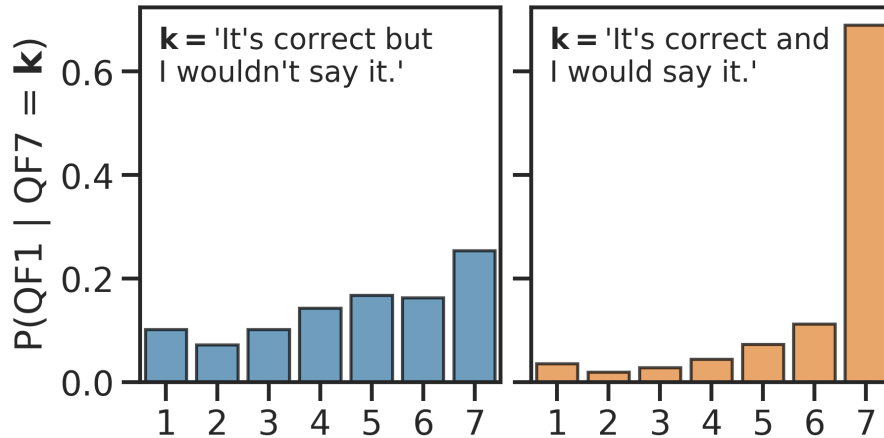


Figure 4: Distribution of QF1 responses for sentences rated by the same speaker in both QF7 and QF1.

4.4 From variability to a common scale

Taken together, these model-free measures establish that protocols are not neutral instruments. They expose three faces of the same protocol-induced variability: (i) speakers converge more under some protocols than others; (ii) a reference translation shifts the sentence judgment; and (iii) a format that splits correctness from use exposes spread that grammaticality alone does not explain. What these measures cannot tell us is how much of this variability survives into the quantity we ultimately care about: the relative acceptability of the predicate-frame pairs we set out to study. Answering that requires placing every protocol on a single, shared acceptability scale, so that a judgment given under one format and modality can be compared with a judgment given under another. We build such a scale in the next section.

5 Synthesizing acceptability judgments across question formats

Section 4 established, using only model-free measures, that protocols differ in the variability they produce. We now aim to assess how much of that variability survives into our inferences about acceptability. We do this with a statistical model that synthesizes the data collected under all protocols into (i) a single acceptability value for each expression; and (ii) a mapping, for each protocol, from that value to the response scale that protocol uses—yielding a proxy for the ‘ground truth’ acceptability of each expression against which we can compare the acceptability we infer under any protocol alone.

5.1 Rationale

Why do we need such a model in the first place? Consider Figure 5, which shows the distribution of responses in each question format. The question formats produce visibly different distributions. Some of this difference is plausibly due to the question format itself—its prompt and its scale-point labels—and possibly to how particular participants use a given scale. But we cannot read a difference in distributions as a difference between the scales, because the question formats were not all applied to the same set of sentences. As described in Section 3, the sentences in each survey were stratified by predicted ac-

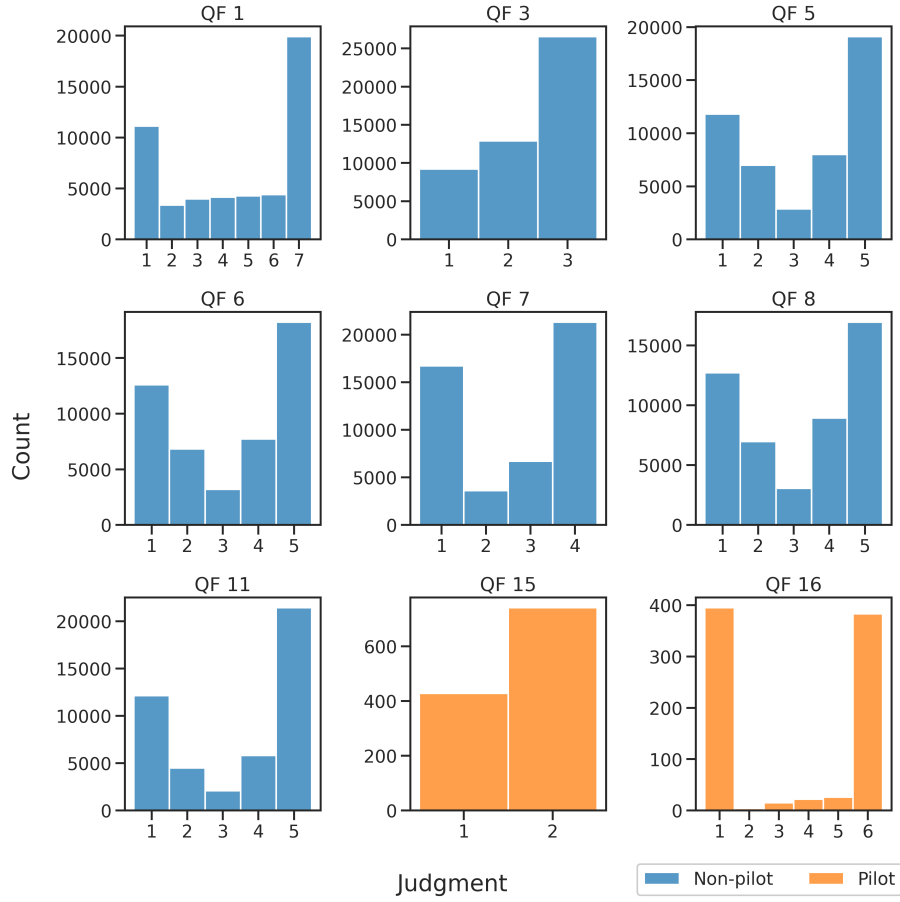


Figure 5: Distributions of responses for each question format.

ceptability and drawn per survey, so a format’s distribution could differ simply because it was applied to more- or less-acceptable sentences. To synthesize the information about acceptability that each format provides, then, we need a model that separates two things: each expression’s underlying, format-independent acceptability, and the mapping of that acceptability onto each format’s response scale. We also need that model to let particular participants differ in how they interpret a format’s prompt and its scale-point labels. The next section lays that model out.

5.2 Model structure

Acceptability judgments, which we operationalize here as particular responses by particular speakers using a particular response instrument, are a product of multiple factors. One such factor—hopefully an important one—is the underlying acceptability of the expression being judged. Another factor is the way the particular speaker maps that underlying acceptability to the scale associated with the question format presented to them. We refer to the underlying acceptability of some expression e as acc_e and the mapping speaker s uses to produce a response using response scale for question format f as respond_s^f . Under these assumptions, we model the n^{th} judgment in our dataset as in (24).

$$(24) \quad \text{judgment}_n = \text{respond}_{\text{speaker}(n)}^{\text{scale}(n)}(\text{acceptability}_{\text{expression}(n)})$$

where $\text{expression}(n)$ tells us the identity of the expression being judged; $\text{scale}(n)$ tells us which scale was used; and $\text{speaker}(n)$ tells us which speaker gave the judgment.

A major assumption of this model is that each expression has a single, speaker-independent acceptability. This assumption is almost certainly false: different idiolects almost certainly vary with respect to the acceptability they assign to at least some expressions; but we take it to be a useful and largely innocent idealization for the purposes of this study—and one that is common (if implicit) in much of the literature.

Acceptability is itself a product of various factors. One such factor—again, hopefully an important one—is grammaticality. But there are presumably others, such as the plausibility of the situation described by a particular expression.

Because our study manipulates the predicate and embedded clause, we assume the interaction of the two is the main driver of grammaticality for these expressions. This underlying grammaticality can itself be modulated by noise due to the particular expression’s form and content. We specifically assume that such noise only degrades acceptability. These assumptions are schematized in (25).

$$(25) \quad \text{acceptability}_e = \text{grammaticality}(\text{predicate}(e), \text{frame}(e)) - \text{degradation}(e)$$

where $\text{predicate}(e)$ tells us the identity of the embedding predicate in the expression; $\text{frame}(e)$ tells us the structure of the embedded clause in the expression; and $\text{degradation}(e)$ represents degradation due to all non-grammatical drivers of acceptability associated with the expression, such as the plausibility of its contents.

In the remainder of the paper, we model acceptability and grammaticality as continuous quantities (Keller 2000; Sorace & Keller 2005; Sprouse 2007; Lau et al. 2017; Sprouse et al. 2018), represented by real numbers. We do not intend this modeling assumption as a commitment to either acceptability or grammaticality being continuously valued properties. We make it merely for the sake of expediency, since modeling either as discrete would require substantial additional non-standard assumptions in the models we use that are not relevant to our goals.

To implement the above model, we work in a generalized linear mixed modeling (GLMM) framework (Gelman & Hill 2006; Bates et al. 2015) that is completely standard, with two exceptions. It is standard in the sense that we model acceptability, as schematized in (25), as an affine equation, as in (26).

$$(26) \quad \text{acceptability}_e = \beta_{\text{predicate}(e)} + \beta_{\text{frame}(e)} + \beta_{\text{predicate}(e) \times \text{frame}(e)} - d_e$$

where β terms are real-valued *fixed effects* and the d terms are real-valued *by-expression random intercepts*.

The first place we diverge from a standard GLMM is in assuming that the random intercepts are non-negative, thereby capturing the idea that non-grammatical effects only degrade acceptability.²

The results we report below do not hinge at all on this assumption: the pattern is nearly identical using a standard GLMM, wherein they can be either positive or negative. We

² To implement this idea, we constrain the d terms (the random intercepts) to be strictly negative using a softplus transform.

take this approach mainly to better align with what we take to be standard informal assumptions in the literature—i.e. a predicate is *grammatical* in a syntactic context if there is some *acceptable* expression instantiating that predicate in that context. In contrast, if we used a more standard GLMM grammaticality would correspond roughly to an average over expression acceptabilities (rather than to what amounts roughly to a max here).

To implement (24), we use a set of ordered logistic link functions, which is the standard for modeling totally ordered, discrete response scales (Bürkner & Vuorre 2019).³ This model assumes (i) that each participant s partitions the real line into a number of intervals $(-\infty, c_{p1}), (c_{p1}, c_{p2}), \dots, (c_{p(k-1)}, \infty)$ equal to the number of response values k they are provided; and (ii) when they are asked to judge an expression with acceptability a , they give the response corresponding to the interval into which $a + \epsilon$ falls, where ϵ is a noise term that simply represents all unexplainable randomness in the judgments (including, e.g., participants’ uncertainty about the true acceptability, but also things as mundane as errors in choosing the intended response). The result is that the response value corresponding to the bin into which a itself falls will have highest probability; but, depending on the level of noise and size of the intervals, adjacent values will also have some amount of probability. We assume that each participant has a unique set of interval boundaries $c_{s1}, \dots, c_{s(k-1)}$ (*cutpoints*) for each response scale, which allows us to account for differences in the way participants use the scale—e.g., some preferring to always respond with endpoints, with others using the scale more uniformly.

As mentioned, this setup is completely standard for tasks involving ordinal scales. Where we diverge from the standard approach is in assuming that each participant has a different set of cutpoints for each question format’s response scale. This assumption is what allows us to synthesize acceptability from multiple, otherwise incomparable response scales—thereby supporting the main aim of this paper—and, conversely, to compare how participants map the same underlying acceptability to different response scales.

5.3 Model fitting

Fitting this model to the full data gives us our best estimate of each expression’s acceptability. The assumption here is that, from the data we describe in Section 3, we have sufficient information to pinpoint acceptability with high enough precision to serve as a reasonable ground truth.

The same setup also lets us focus on specific parts of the dataset—e.g. only elicitation data collected using a single question format—and compare the acceptability we estimate from such a subset with the acceptability we estimate from the full data. This lets us simulate situations in which we have scarcer data than we actually do, and it is the comparison on which we base our ultimate recommendations.

We report the comparisons these subsets support in Section 6. For all such comparisons, we employ the same procedure. We fit our models using the PyMC Bayesian modeling framework and Python package (Abril-Pla et al. 2023), with 4 chains and 1000 draws (with 1000 draws for warmup) of Hamiltonian Monte Carlo. Sampling is performed on one RTX 5000 GPU using the NumPyro library (Phan et al. 2019) with the JAX backend (Bradbury et al. 2018). In most cases, the relevant model parameters pass modern fit

³ Not all question formats’ response scales are natively totally ordered: some response scales, such as the ones associated with QFs 3, 7, and 11, are likely more naturally thought of as partially ordered. For the purposes of this model, we mapped responses from each question format onto an ordinal scale as described in Section A.

diagnostic thresholds as suggested by [Vehtari et al. \(2021\)](#). However, 7 models have at least one parameter with $\hat{R} > 1.01$ and with an effective sample size (ESS) < 400 , all of which were models fit to the simulated scarce data subsets. This is likely indicative of the issue that there is genuinely low information for some parameters given very low data.⁴

5.4 Qualitative validation of predicate $\times$ frame compatibility

Figure 6 visualizes the predicted compatibility (posterior mean) of each verb in each frame when fit to the full dataset as calculated from only the fixed effects ($\beta_{\text{predicate}} + \beta_{\text{frame}} + \beta_{\text{predicate} \times \text{frame}}$) removing the random intercepts (degradation terms). As a reminder, the values themselves are continuous and only have a relative interpretation. But to make those values easier to interpret, we have projected the predicted compatibility back onto the response scale for question format 1, using the mean cutpoints across participants. Thus, the gradient represents the acceptability on a scale from 1-7, with the center of the gradient (white) at the midpoint (4). Cross-hatching indicates predicate-clause pairs for which we do not have data.

As noted earlier, we are not aware of any previous study of Zulu embedding that systematically investigates predicate–frame compatibility across this range of predicates and all six complement types. Consequently, we cannot evaluate our findings in terms of replication of an established body of results. Nevertheless, our data confirm a number of judgments reported in the previous literature (e.g. [Slattery 1981](#); [Ziervogel et al. 1985](#); [Van der Spuy 2001](#); [Buell 2005](#); [Zeller 2006](#); [Halpert 2015](#)), including the observation that *fanele* ‘must’ and other raising predicates select subjunctive clauses, and that subject control predicates such as *zama* ‘try’ and *funa* ‘want’ select infinitival complements. The results also replicate the observation, reported for example by [Slattery 1981](#), that the so-called adverbial auxiliaries preferentially combine with small subjunctive complements.

A more systematic investigation of embedding patterns has been carried out for Zimbabwean Ndebele, a language mutually intelligible with Zulu ([Pietraszko 2017; 2020](#)). The Zulu data reveal some of the same correlations observed in Ndebele. For example, for most predicates, compatibility with infinitival complements closely tracks compatibility with subjunctive CPs, and it is rare for a predicate to combine with both indicative CPs and small subjunctives. These correlations require further analysis and interpretation, which we leave for future work. One finding that already merits attention, however, is that infinitives and gerunds exhibit clearly distinct distributions, supporting the hypothesis that they are distinct clause types in Zulu. To our knowledge, this distinction has not previously been demonstrated on the basis of a systematic distributional comparison.

5.5 How mappings differ across question formats

Because every speaker’s cutpoints live on the same continuous acceptability scale, they give us a direct way to compare what different question formats do with that scale (Figure 7). To read the figure: for QF1, the leftmost cutpoint is the acceptability at which the median speaker becomes more likely to respond 2/7 than 1/7.

We compare formats along two dimensions. The first is *response-scale type*—numerical with described endpoints (QF1, a 1–7 scale) versus fully descriptive (QFs 5, 6, 8). QF1’s

⁴ For these models, the average maximum $\hat{R}$ is 1.017 (average 95% = 1.008), and the average minimum ESS is 63 (average 5% = 148). The low maximum $\hat{R}$ suggests that the models actually converge reasonably well, and a minimum ESS of around 60 is likely sufficient for our inferences ([Gelman et al. 2003](#): p.287).

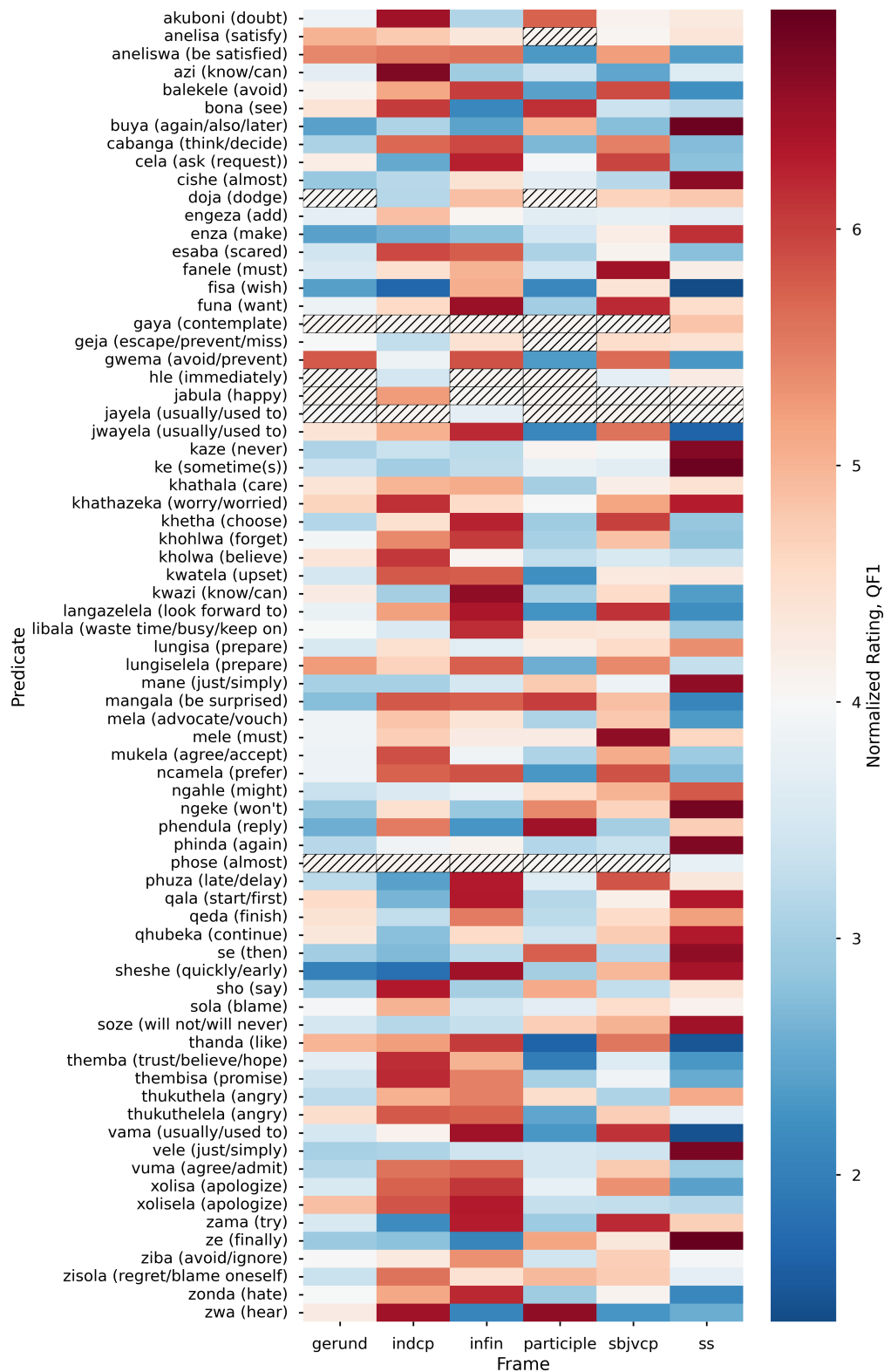


Figure 6: Full-model estimated predicate × frame compatibility. Hatched cells indicate predicate × frame pairs that were not tested in this study. Since sentence random effects were constrained to be negative, these represent the hypothetical ‘max’ acceptability of a particular predicate × frame instantiation.

cutpoints are spread fairly evenly around 0, whereas the descriptive formats' cutpoints are not, and span a different range (Figure 7); this suggests that QF1 yields, on average, a more stable mapping from acceptability to response than the descriptive formats do. The middling descriptive response ("I'm not sure") maps to only a narrow band of acceptability (QF5: $[-0.13, 0.32]$, width 0.45), while "Probably"/"Probably not" cover a wider band (QF5: $[-1.28, -0.13]$, width 1.15). The top descriptive cutpoint is higher than QF1's (QF5: 1.50 vs. 1.42) and the bottom descriptive cutpoint is not as low as QF1's (QF5: -1.28 vs. -1.41)—i.e. "No problem" demands higher acceptability than a 7/7, and "Definitely not" does not require acceptability as low as a 1/7.

The second dimension is *perspective on acceptability*—sentence-centered (QF1), participant-centered (QFs 5, 6), and population-centered (QF8). The cutpoints for QFs 5, 6, and 8 are qualitatively similar, differing meaningfully only at the highest response level: QF5 (1.50), QF6 (1.49), and QF8 (1.61) all set a higher bar there than QF1 (1.42), and QF8 (population-centered) sets the highest bar of the three descriptive formats. This suggests that speakers are most stringent about their top response when judging for a generic Zulu speaker, less so when judging their own usage, and least when judging the sentence itself. This is an interesting result. One might have expected the population-centered prompt to encourage greater leniency, since speakers are likely aware that not all native speakers judge every expression identically, making it sufficient that *some* speakers would find a sentence perfectly acceptable. Instead, the opposite pattern emerges. One possible interpretation is that speakers implicitly treat their own judgments as somewhat more permissive than those they attribute to the average Zulu speaker.

These mapping differences are exactly what the model absorbs when it places all formats on one scale, which is what lets the later analyses treat judgments from different protocols as comparable.

6 Estimating predicate-frame acceptability from data subsets

With the model of Section 5 placing every protocol on a common acceptability scale, we can now ask how our data-collection choices feed into the estimates we ultimately care about—the relative acceptability of the predicate-frame pairs we study, which we refer to throughout as *predicate-frame acceptability*. We do this by fitting the same model to subsets of the data and comparing the predicate-frame acceptability each subset supports with the predicate-frame acceptability the full data support. The fitting procedure is the one described in Section 5.3; here we report the comparisons.

6.1 Question formats and translations do not bias predicate-frame estimates

In Section 4 we found that pairing a Zulu sentence with a reference translation shifts the raw sentence judgment, especially in the presence of non-target translation. That shift is a fact about the raw responses. Whether it also biases our model-based estimates of predicate-frame acceptability is a separate question, and it is the one we take up here.

We run two checks. First, we fit the model to each question format on its own and compare the predicate-frame acceptability it infers with those from the full-data fit. Then we fit it to subsets from which the non-target translations have been removed and again compare with the full-data fit.

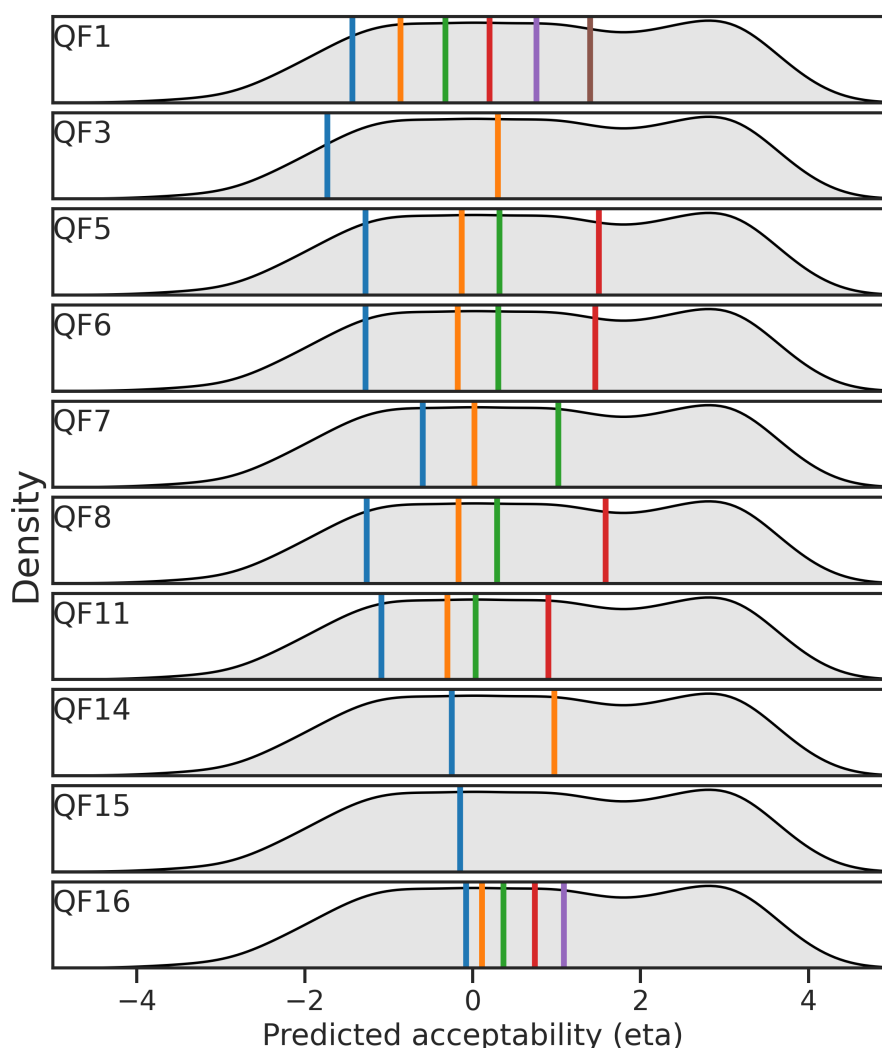


Figure 7: Model-inferred median cutpoints across speakers. In gray: distribution of sentences across predicted latent acceptability.

Both checks point the same way. All single-format subset models reach high concordance with the full-data model, including those that still contain non-target translations (Figure 8). The full model and the model with only target translations correlate nearly perfectly (Spearman $\rho = 0.95$; Figure 9). Thus the non-target translation effect observed in the raw data does not meaningfully bias our estimates of predicate-frame acceptability—the model absorbs it before it reaches the quantity we report. The fact that all of the single-format models reach a similarly high correlation with the full-data model also indicates that all of the question formats perform quite well in estimating the predicate-frame effect.

6.2 Limited data suffices to recover predicate-frame acceptability

We collected predicate-frame judgments under many protocols. Most fieldwork cannot, especially fieldwork on lower-resource languages, where the depth of judgment data assembled here is rarely within reach. So we ask two practical questions: how little data suffices to recover the predicate-frame acceptabilities that this larger study supports

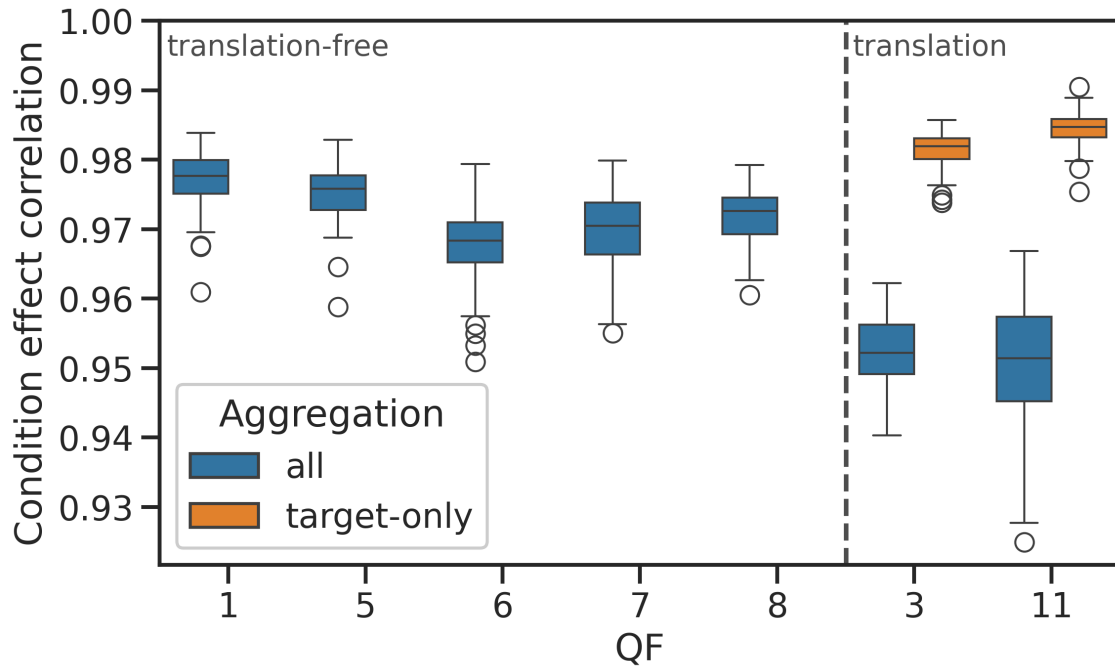


Figure 8: Distribution of bootstrapped Spearman correlations. Correlations are computed using estimated predicate-frame acceptability, between models fit to single question formats and a model fit to the full data. For translation-based question formats, we fit two models—one to all the data, and one with only target translations included.

(Myers 2009; Mahowald et al. 2016; Sprouse & Almeida 2017), and how far online studies can replicate what elicitation yields.

We answer them as before, by comparing the predicate-frame acceptabilities from subset models with those from the full-data model. We fit the model to online responses to QF1 at 50 and 90 speakers (each at 100 and 200 sentences per speaker). Due to limited amounts of elicitation data, we also fit the model to each of online-only and elicitation-only responses to QF1 at 10 speakers (each at 20, 40, and 50 sentences per speaker). For each subset we fit 5 models, each on a different random draw of speakers, so that a single comparison reflects a distribution over draws rather than one lucky or unlucky sample.

Even a model fit to 10 speakers with 50 sentences per speaker already yields a good correlation of predicate-frame acceptability with the full-data model (Figure 10, Spearman rho 95% interval = [0.72, 0.82]). Adding speakers helps along two dimensions at once: it raises the correlation with the full model and it shrinks the variance of that correlation across draws.

The upshot for practice is encouraging but requires qualification. Online studies can recover the same predicate-frame acceptabilities that elicitation does, but the protocol-induced variability established in Section 4 means it takes roughly three times as much online data to get there—and, extrapolating the subset curves to the highest levels of precision, as much as ten times.

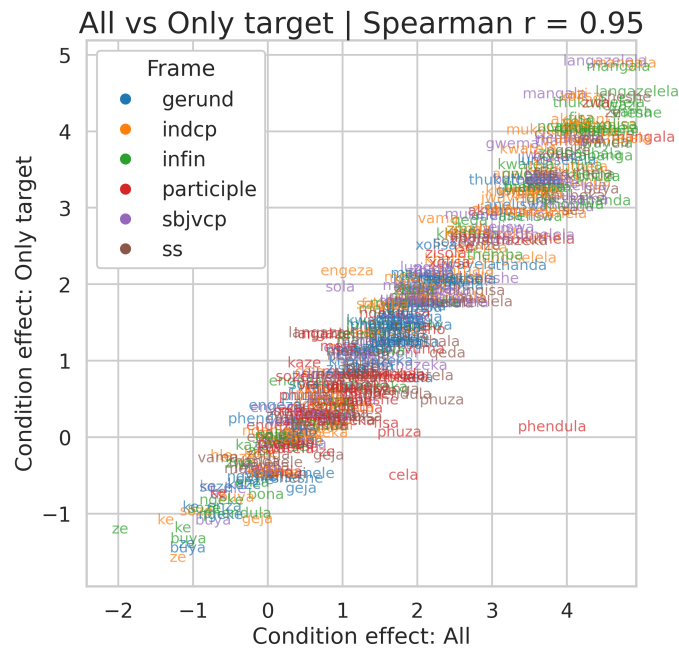


Figure 9: Predicate-frame acceptability in a model fit to the full data (x) versus a model fit to the full data with confounds removed (y).

7 Discussion

When working on a relatively unfamiliar language, researchers have to make a variety of choices about the protocol they will use to collect their data. Each choice has trade-offs.

7.1 Modality and setting

We find that elicitation allows one to obtain precise predicate-frame acceptability estimates with the least data. We thus recommend using elicitation over unguided online surveys whenever it suffices to achieve the desired scale of the study.

Among the elicitation settings we investigated, group elicitation yielded more agreement among speakers in their judgments. The apparent advantage of group elicitation naturally raises the question of why it arises. One possibility is that discussion helps speakers refine their judgments by drawing attention to relevant linguistic contrasts or interpretations, including ones they did not think of themselves. Another is that the increased agreement simply reflects conformity—speakers converging on a shared answer through discussion—rather than being an indication of gained accuracy.

Our design guards against the second possibility in part, as participants recorded their judgments independently after the discussion rather than agreeing on a single group response. This independence makes pure conformity an incomplete account of the group effect, but it cannot rule conformity out: a speaker may still privately adjust toward the group after hearing it, and our data do not let us separate such private adjustment from a *bona fide* sharpening of the judgment. With this caveat, we suggest that there is indeed improved efficiency from using group elicitation.

Finally, we note that the group effect is based on a small pilot sample (eight speakers) and should therefore be interpreted with caution. But even on a conservative interpreta-

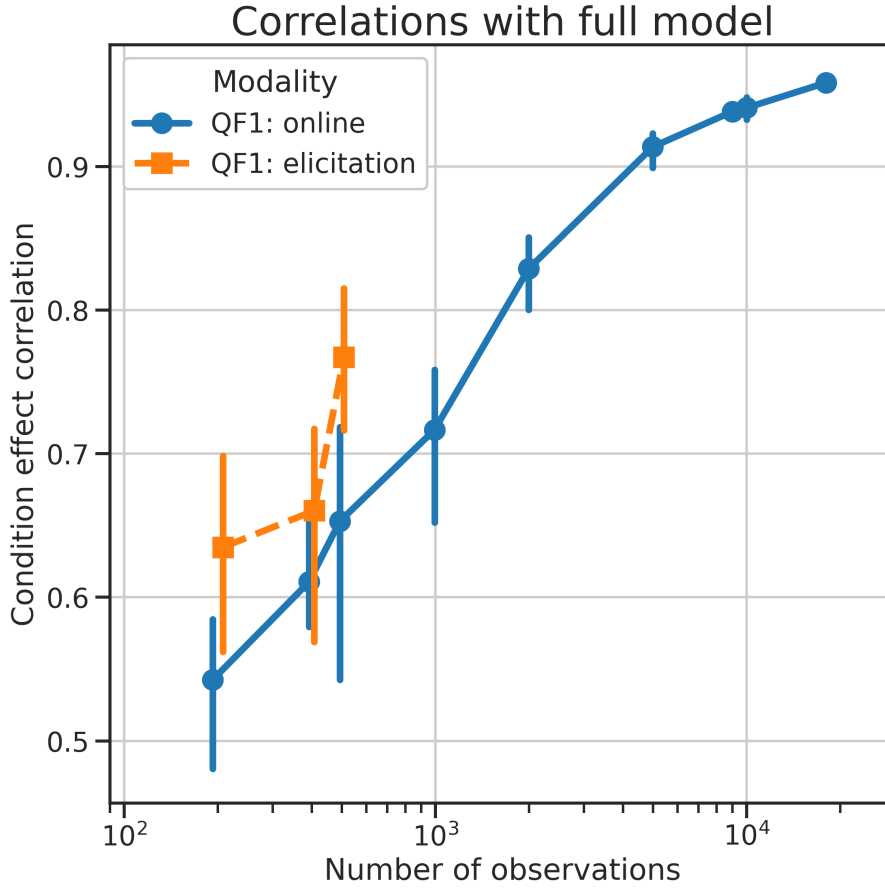


Figure 10: Spearman correlations of inferred predicate-frame acceptability. Correlations computed between models fit to varying subsets of data against the full model. Errorbars represent bootstrapped 95% percentile intervals.

tion, group elicitation performed at least as well as individual elicitation, while allowing substantially more judgments to be collected in the same amount of time.

Nonetheless, as discussed in Section 1, elicitation is not always feasible at the scale required by a given research question. We find that unguided online surveys provide an effective alternative when the amount of data required exceeds what can realistically be collected through elicitation. They produce the same estimates of predicate-frame acceptability as elicitation; they just require substantially more data to do so: at least three times as much, and potentially as much as ten times, to obtain similar estimates. Thus, insofar as one has access to sufficient numbers of participants and the ability to deploy online surveys to those participants, our conclusion is that online studies can be used successfully (possibly in combination with elicitation) when scale is necessary for the question at hand.

7.2 Question format

Which subset of question formats to choose in any particular case depends heavily on which areas of the acceptability scale are of most interest. We suggest that QF1 is a reasonable default because its cutpoints are distributed relatively evenly across the acceptability scale. This even spacing means that, unless the underlying acceptability of a

target set of items is very heavily concentrated within a particular area of the scale, QF1 will tend to allow a researcher to more precisely identify an expression's acceptability than most other formats. QF1 contrasts with the fully descriptive formats (QFs 5, 6, and 8), whose cutpoints are unevenly spaced and whose highest response category requires a higher level of acceptability than the highest numerical rating in QF1.

At the same time, QFs 5, 6, and 8 performed just as well as QF1 in producing estimates of predicate-frame acceptability with the sample we focused on and with the sample sizes we had. Their principal advantage is that the descriptive response labels are easier for researchers to interpret than unlabeled numerical ratings, which reflect not just well-formedness but other dimensions as well. We therefore recommend a descriptive question format like this when ease of interpretation is a priority and fine-grained distinctions across the acceptability scale are less critical.

Relatedly, our results show that separating judgments of well-formedness from judgments of likely use reveals that a substantial portion of the variation in unlabeled numerical ratings reflects use-related considerations rather than well-formedness itself. We therefore recommend including a question format that distinguishes these concepts—e.g. our QF7—whenever distinguishing these two components of speakers' judgments is important for the research question.

A final recommendation concerns the translation-relative formats (QF3, QF11), which pair each Zulu sentence with an English translation. We recommend using these formats wherever possible, because the translation produces measurably more precise estimates in the presence of parsing confounds: when a sentence has more than one parse and we care only about the target parse, the translation biases the speaker toward that parse, so the judgment we collect is a judgment of the structure we meant to test. Translation-free formats cannot do this biasing by definition: facing an ambiguous sentence, a speaker is free to judge it on whichever parse they land on, possibly a non-target one.

One might thus have worried that translation-free data would be confounded by non-target parses—participants judging ambiguous sentences on readings we did not intend. Surprisingly, however, we found that translation-free data does not appear to be confounded—at least for the kind of phenomena we were investigating and for our protocol design. Translation-free judgments correlate more closely with the judgments speakers gave the target translation alone than with an aggregate—a median or a max—over target and non-target translations (Section 4), which suggests that translation-free responses were judgments against the target parse.

We suspect that this pattern arises as a result of participants implicitly identifying the target parse we had in mind, generalizing from the many other sentences they judged that unambiguously had the target clause-embedding structure. This robustness may thus hold only when a study has a consistent focus (such as clause-embedding structures) that participants can pick up on. Indeed, in contexts where the researcher aims to conceal the manipulation of interest, we might expect parsing confounds to have stronger effects on participants' judgments.

7.3 The benefits of an inferred acceptability model

The discussion above points toward mixing and matching protocols: using elicitation in one place and unguided online surveys in another, and varying the question format with the question one wants to answer. But a researcher who mixes protocols this freely

faces a problem the recommendations themselves do not solve. Without a way to bring data collected under different protocols onto one scale, mixing and matching yields a heap of judgments that needs to be synthesized into a single measure of acceptability. A model that places otherwise incomparable protocols on a common acceptability scale, such as the one we present in Section 5, supports this synthesis, because it separates predicate–frame acceptability from the format- and protocol-specific distortions that the raw judgments carry.

The model recovers consistent predicate–frame acceptabilities even from thin data, with models fit to a single question format, or to as few as ten speakers, agreeing closely with the full-data model, and every single-format model concurring with it (Section 6). It also absorbs protocol-specific distortions: the parsing bias by which a reference translation shifts the raw judgment of a sentence does not survive into the model-based predicate–frame acceptability estimates (Section 6). Most directly to the point, elicitation and unguided online surveys, which differ sharply in the protocol-induced variability they produce (Section 4), yield the same predicate–frame acceptabilities once the model places them on a common scale (Section 6). We therefore recommend that researchers who must combine data collected under different protocols use a model of this kind.

8 Conclusion

Taken together, these recommendations mean that the inverse relationship between language familiarity and required experimental control need not force an all-or-nothing choice between elicitation and unguided survey. A researcher who needs both scale and precision can collect the bulk of their data through unguided surveys, anchor it with a smaller block of elicitation, ideally group elicitation, and let a common-scale model reconcile the two. What that model enables is the freedom to let the question dictate the scale and the budget the mix.

Abbreviations

We use the Leipzig glossing convention with the following additions: 1SG.S – first person singular subject agreement (etc.), 1.S – class one subject agreement (etc.), AU – augment vowel, DSJ – disjoint, GER – gerund.

Data availability

Data and code used for this analysis are available at https://osf.io/cwp9y/overview?view_only=42355393dd714e9e9de7f188827ce32b.

Ethics and consent

All research procedures were approved by the Institutional Review Board at the University of Rochester and the Humanities and Social Sciences Research Ethics Committee at the University of KwaZulu-Natal.

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Competing interests

The authors have no competing interests to declare.

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A Question Format Coding

Format 1

Is this a good isiZulu sentence?

[Zulu sentence]

Response option	Score
Perfectly normal	7
Intermediate points on a 7-point scale	2–6
Completely broken	1

Format 5

Is this something you might say?

[Zulu sentence]

Response option	Score
Definitely yes	5
Probably	4
I'm not sure	3
Probably not	2
Definitely not	1

Format 6

Can you imagine yourself saying:

[Zulu sentence]

Response option	Score
No problem	5
Probably	4
I'm not sure	3
Probably not	2
Definitely not	1

Format 7

Which statement below best describes the sentence:

[Zulu sentence]

Response option	Score
It's correct and I can imagine saying it.	4
It's correct but I wouldn't say it.	3
I'm not sure if it's correct or broken.	2
It's broken.	1

Note: This question format could alternatively be coded with both responses beginning with 'It's correct' as corresponding to the same (top) value. We chose to differentiate them following our findings in Section 4.3 that the would-use sentences are rated significantly higher than the wouldn't-use.

Format 8

Would a native speaker of isiZulu say:

[Zulu sentence]

Response option	Score
No problem	5
Probably	4
I'm not sure	3
Probably not	2
Definitely not	1

Format 3

Which statement is true about the sentence:

[Zulu sentence]

[English sentence]

Response option	Score	English Score
The Zulu sentence is good and the translation is ok.	3	1
The Zulu sentence is good but the translation is incorrect.	3	0
The Zulu sentence sounds a bit odd but the translation is ok.	2	1
The Zulu sentence sounds a bit odd and the translation is incorrect.	2	0
The Zulu sentence is broken.	1	–

Note: Since the judgment on the Zulu sentence is independent of the judgment on the translation, we code the Zulu judgment independently (e.g. both responses with ‘The Zulu sentence is good’ are mapped to the top level).

Format 11

Is this a good isiZulu sentence?

[Zulu sentence]

[English sentence]

Response option	Score	English Score
It’s perfectly normal.	5	1
I’m not sure if I would say it, but it sounds correct.	4	1
I’m not sure if it’s correct or broken.	3	1
I think it’s broken.	2	1
It’s definitely broken.	1	1
It’s not broken but it means something different.	–	0

Note: Since the bottom response (‘It’s not broken but it means something different’) is ambiguous as to how not broken the Zulu sentence is, we chose to drop this response level when analyzing the Zulu acceptability.

B Predictive model for sentence stratification

As described in Section 3.3, we used a version of the synthesizing model to stratify the sentences tested in the online studies and thereby encourage participants to calibrate their scale use to the distribution of acceptabilities for sentences found in our materials. In order to generalize to unseen sentences, we included sentence-level predictors from a pre-trained transformer masked language model. Masked language modeling, first introduced by Devlin et al. (2019), is a self-supervised task in which a model learns representations of text by training on predicting the masked tokens (i.e. hidden words) in a string.

B.1 Language model details

We compiled five previously published Zulu corpora (Eiselen & Puttkammer 2014; Goldhahn et al. 2012; Spiegler et al. 2010; Wikimedia Foundation 2023; Strassel & Tracey 2016) with a total size of around 16 million words, mainly consisting of web text. We then trained a byte-pair encoding tokenizer (with a vocabulary size of 32,000) and a slightly modified RoBERTa masked language model (Liu et al. 2019) with 8 hidden layers, a batch size of 64, and a learning rate of 1×10^{-4} on the dataset. The model was trained for a total of 3 epochs, after which the validation loss stopped improving.

B.2 Sentence representation

Using the pre-trained model, we computed embeddings for the sentences in the Pilot and Durban datasets. The embeddings were computed as the mean-pooled outputs of the last hidden layer of the language model. To improve efficiency, we applied PCA to the mean-pooled embeddings for each sentence to additionally reduce the embedding from the language models' original 768-dimensional embedding down to 10 dimensions. Finally, these 10-dimensional embeddings and the tokenized length of the sentence were used as additional fixed predictors in the synthesizing model discussed in Section 5 (in addition to the predicate-frame predictors and random intercepts). Length was included as we noticed a small but significant drop-off in acceptability from tokenized length (Spearman $\rho = -0.07$, $p = 0.02$).

This model was fit on the data from the Pilot and Durban studies. Once fit, we generated the language model features (embeddings and length) for each sentence to be tested in the online surveys. We then computed the posterior predictive distribution over acceptability for each sentence. The mean predicted latent acceptability was then binned and used to stratify the sentences for the online surveys.